CareHub — Healthcare Facility Management System

A digital platform for aged care facility management

CSTP2204 - IT DEVELOPMENT PROJECT

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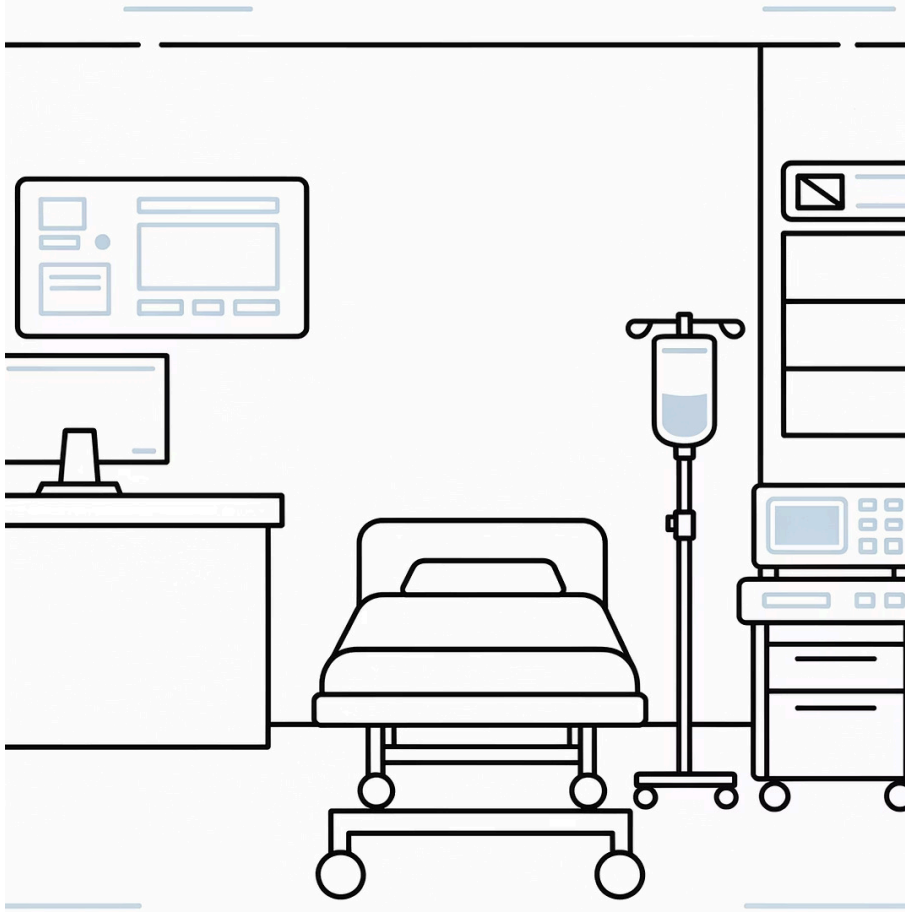


Agenda

What we'll cover today — from architecture to live demo

1. **Project Introduction & Architecture**
   The problem, the solution, and how it's built
2. **Clinical Workflows**
   MAR, medication orders, and inventory management
3. **Web App & AI Features**
   React frontend and Groq-powered clinical tools
4. **Mobile App & Deployment**
   React Native cross-platform app and CI/CD pipeline
5. **Live Demo & Conclusion**
   Hands-on walkthrough, key achievements, and Q&A

What is CareHub?

The Problem

Manual record-keeping in aged care facilities is **error-prone, time-consuming, and disconnected** across staff shifts and departments. Critical medication and observation data is fragmented.

The Solution

A **digital platform** spanning Desktop, Web, and Mobile that manages residents, medications, observations, and staff — with AI-powered clinical decision support built in.

Technology Stack & Architecture

Technology Stack

Backend

ASP.NET Core 8.0 • Entity Framework Core • PostgreSQL 16

Desktop / Web / Mobile

.NET MAUI (Windows) • React + Vite • React Native (Android/iOS)

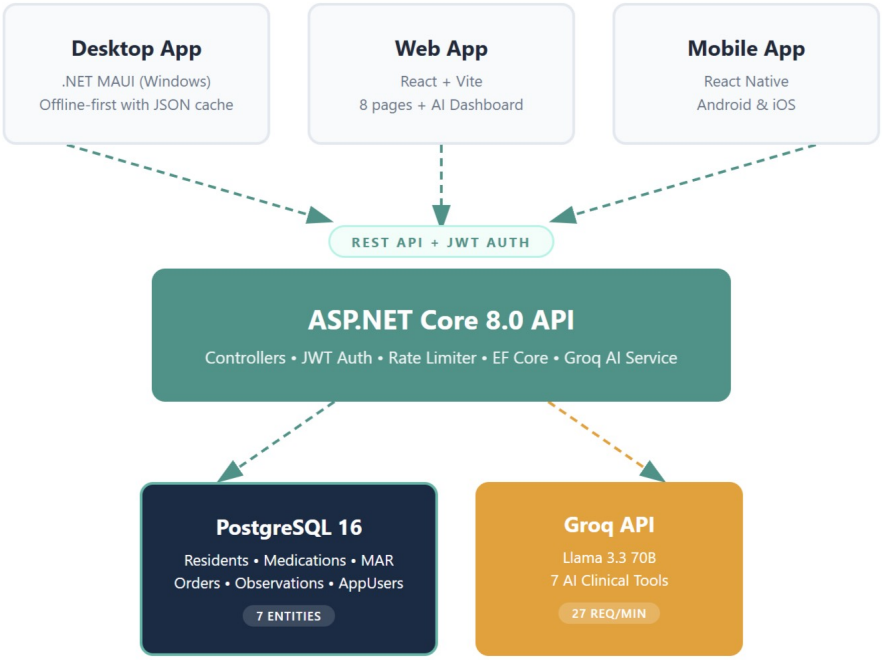
AI & Auth

Groq API • Llama 3.3 70B • JWT Bearer Tokens • BCrypt hashing

Infrastructure

Docker Compose • Railway (API) • Netlify (Web) • GitHub Actions

System Architecture



OFFLINE-FIRST
Desktop caches data locally as JSON, syncs when connected

SYNC QUEUE
ClientRequestId idempotency prevents duplicate entries

JWT + RBAC
4 roles: Admin, Nurse, General CareStaff, Observer

AI INTEGRATION
Rate-limited Groq API with MockAiService fallback

Database & Authentication

Entity Data Models

- Resident
- Medication
- MedicationOrder
- MarEntry
- Observation
- InventoryLedger
- AppUser

Security & Access Control

Role-Based Access

Admin, Staff, CareStaff, Observer each with scoped permissions

JWT Authentication

Bearer token auth with **BCrypt password hashing** for secure credential storage

Realistic Demo Data

Automated **data seeding service** populates all entities with realistic scenarios

Clinical Workflows

Medication Administration (MAR)

Time-based scheduling with configurable daily times. Tracks dose, resident, time, and nurse. Validates against **allergies and duplicate administration**.

Medication Orders

Three-step lifecycle: **Request → Order → Receive**. Full status tracking with timestamps and validation at every transition.

Inventory Management

Batch-level tracking per medication. The **MedicationInventoryLedger** records every stock movement for complete auditability.

CareHub Web Application

Built with **React + Vite** for fast, modern development

Dashboard
overview stats

Residents
list with search & profiles

Observations
vitals and clinical notes log

Medications
catalog with dosage & schedules

Inventory
stock levels and batch tracking

MAR
administration with dose tracking

Orders
order requests with status board

Staff
directory with role management

AI Dashboard
AI powered clinical tools

AI Integration

Powered by Groq API · Llama 3.3 70B, Intelligent rate limiting: **25 requests/min globally, 10 per user** via sliding window, MockAiService fallback ensures offline resilience.

Medication Reference

On-demand drug information lookup

Detect Vital Trends

Trend analysis with plain-language explanations

Report Generation

Structured clinical reports on demand

Shift Handoff Summary

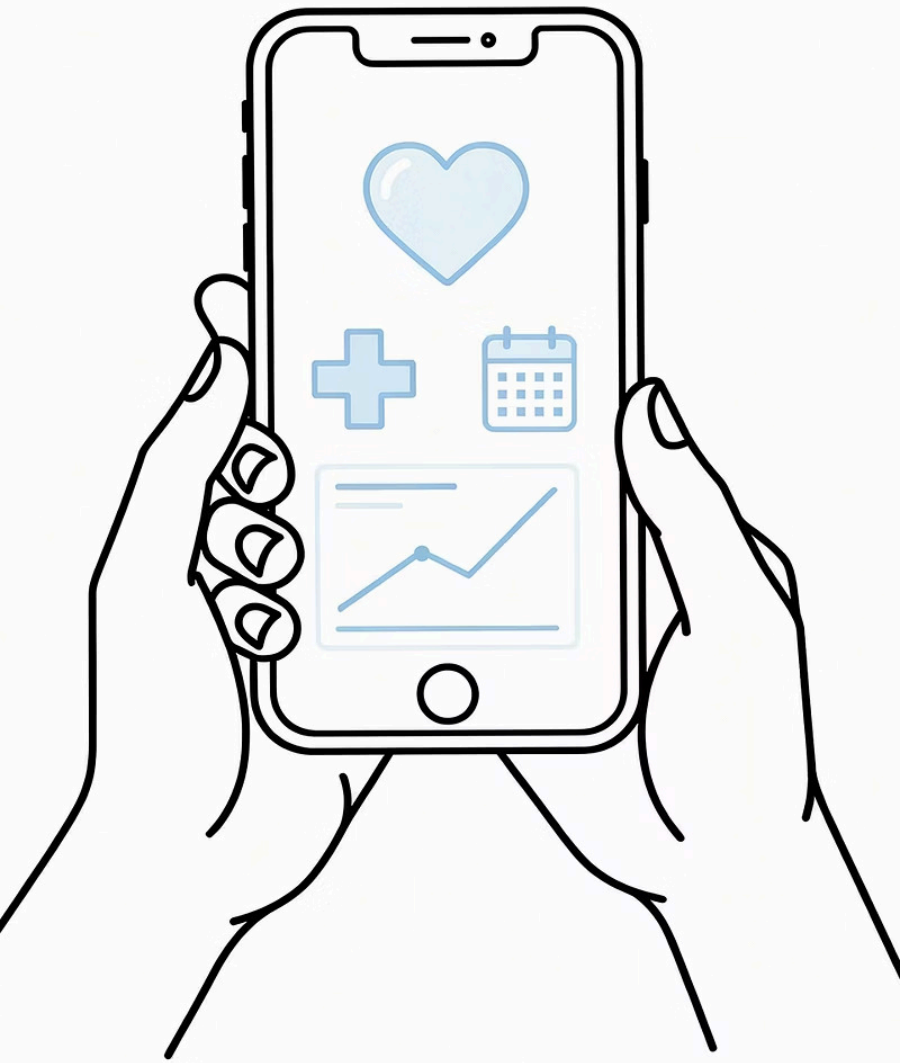
Facility-wide briefing for shift changes

Care Query

Natural language questions about any resident

Resident 24-Hour Summary

Individual clinical narrative report



React Native Mobile App

Cross-Platform

Single codebase for **Android & iOS** via React Native — no duplication, consistent experience

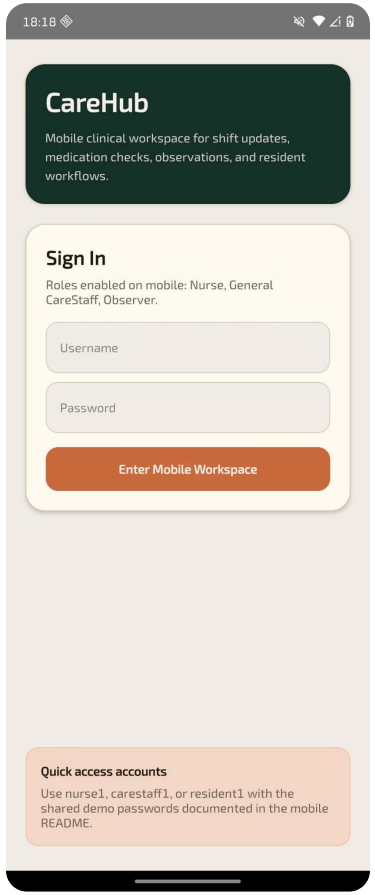
Secure Auth

AuthContext manages JWT tokens, auto-refresh, and all secure API calls

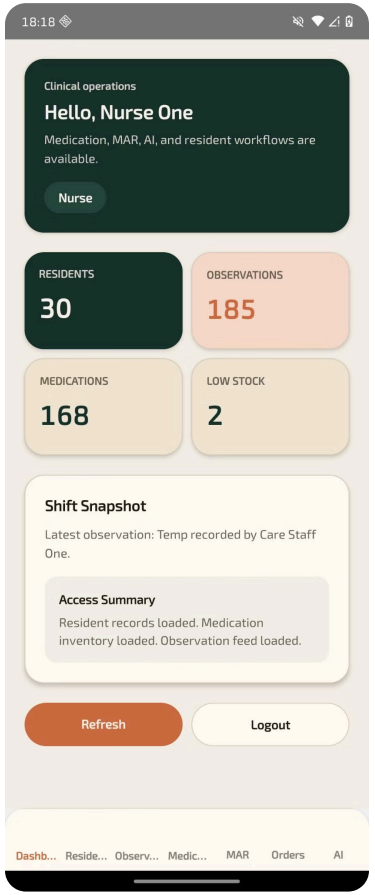
8 Full Screens

Login, Dashboard, Residents, Observations, Medications, MAR, Orders, and AI

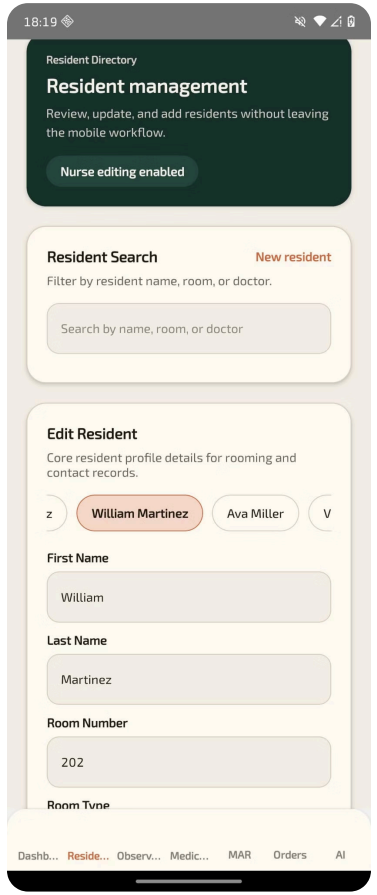
Mobile App Screens



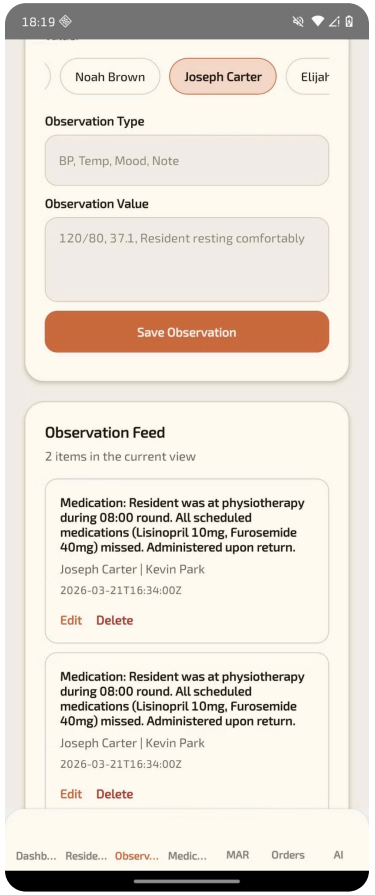
Login



Dashboard

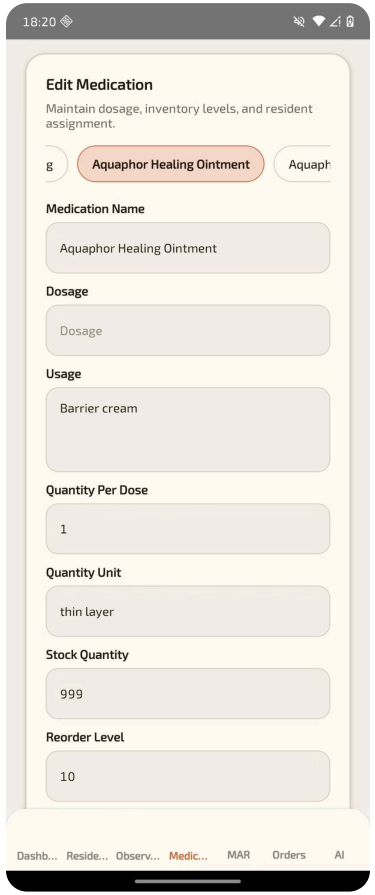


Residents

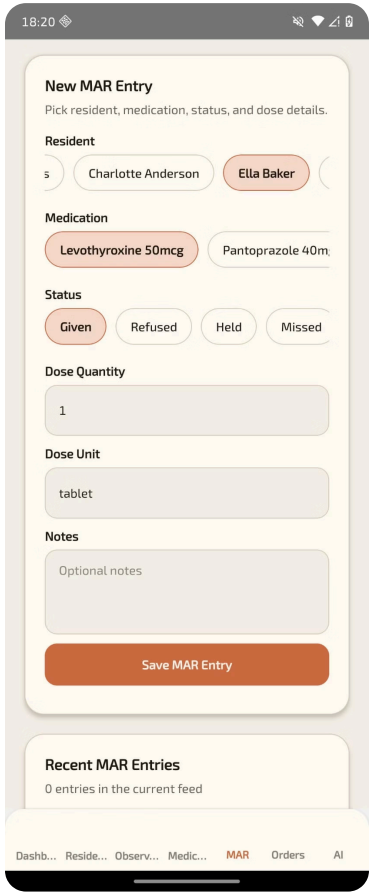


Observations

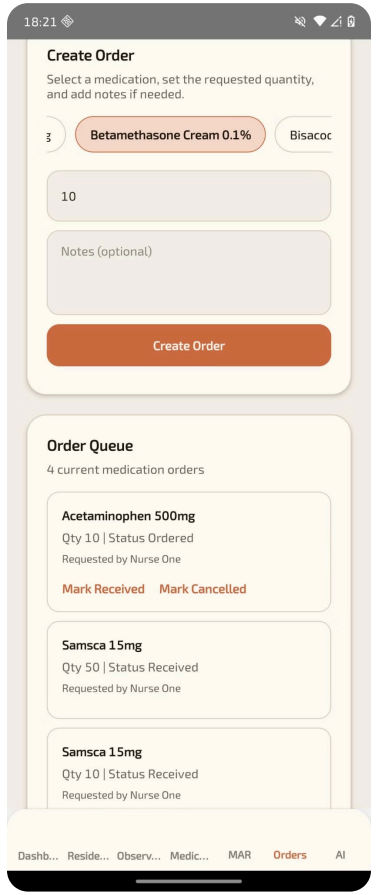
Mobile App Screens



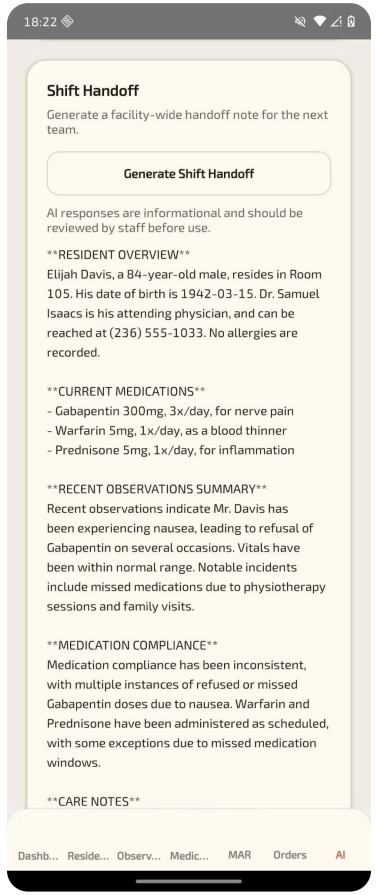
Medication



MAR Entry



Ordering



AI Report

CI/CD & Deployment Pipeline

Every push to the repository triggers an automated build pipeline — ensuring consistent, reproducible deployments from local development through to production.

Docker Compose

local development (API+PostgreSQL)

Railway

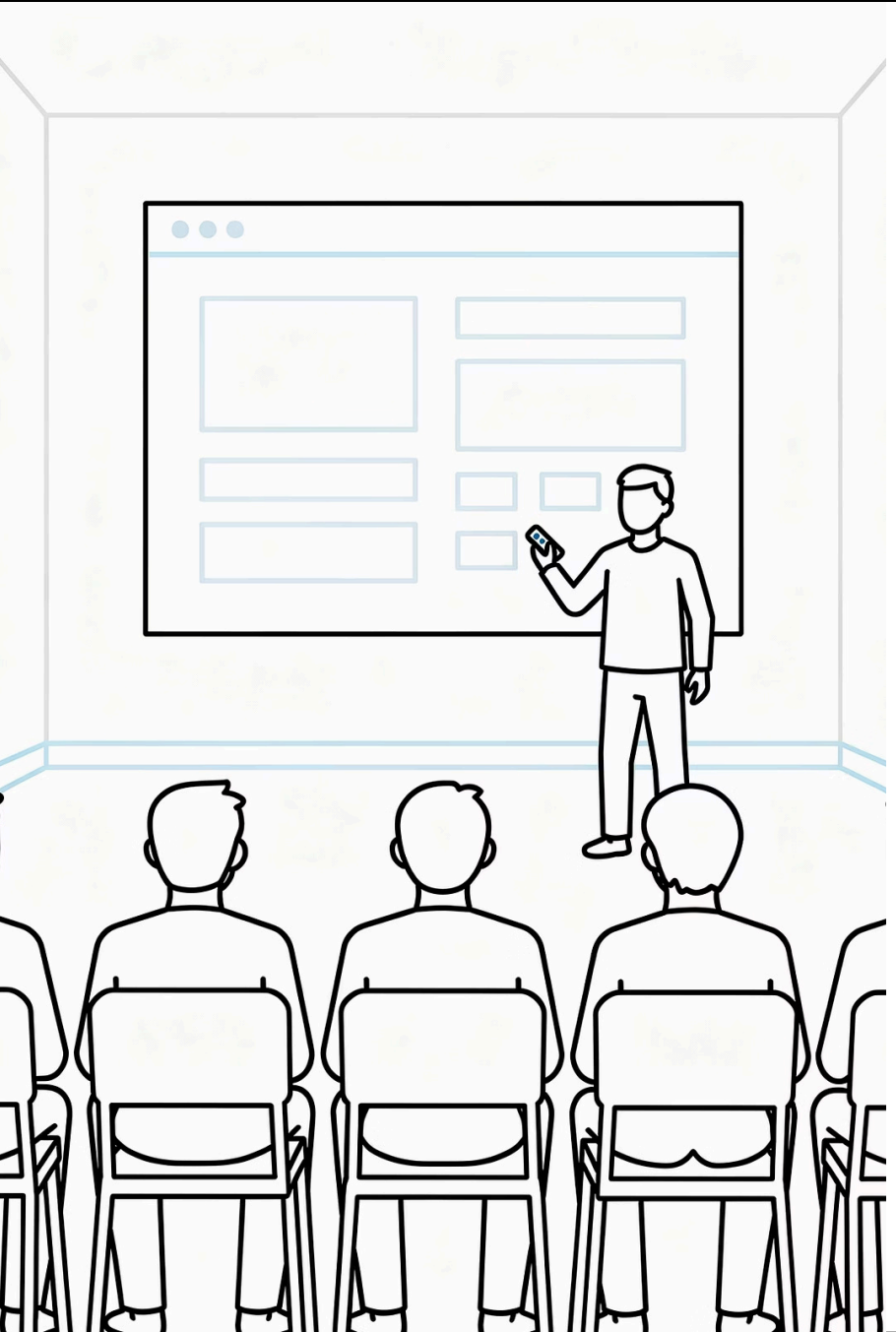
production API hosting with managed PostgreSQL

Netlify

web frontend hosting with auto deploys from Git

GitHub Actions

automated build pipeline on push to master



Live Demo

Desktop App

1. Login with role-based access control
2. Residents page — full CRUD operations
3. Medications & MAR — administer a dose
4. Medication orders — full order lifecycle
5. Observations — add vitals
6. Floor plan view

Web App & AI Dashboard

1. Dashboard overview with live stats
2. AI Dashboard — run Shift Summary live
3. AI Dashboard — run Care Query live
4. AI Dashboard — run Trend Analysis live
5. Role-based access enforcement demo

Summary & Key Achievements

CareHub delivers a complete, production-ready healthcare management platform across every target environment.

Platforms Delivered

Desktop, Web, Mobile

Offline-First Architecture

Reliable sync with ClientRequestId idempotency

AI-Powered

clinical decision support

Complete MAR Workflow

Nurse validation with allergy and duplicate checks

Role-based Access

Role-based access across all platform

Automated Deployment

GitHub Actions, Railway, and Netlify pipeline

Challenges & Solutions

1

Offline Data Sync

Challenge: Mobile and desktop clients must remain functional without connectivity

Solution: ClientRequestId idempotency ensures safe replay of queued requests upon reconnection

2

AI Rate Limits

Challenge: Free-tier Groq API imposes strict request ceilings

Solution: Per-user (10/min) + global (25/min) sliding window rate limiter with graceful MockAiService fallback

3

Cross-Platform Consistency

Challenge: Three client types with different UI frameworks must behave identically

Solution: Shared REST API contract with a service abstraction layer per client platform

Future Roadmap

Real-Time Notifications

Push alerts for medication reminders and critical clinical events

Barcode / QR Scanning

Mobile-based medication verification to reduce administration errors

Advanced Analytics

Interactive charts and trend dashboards for facility-level insights

Family Portal

Secure resident update feeds for family members and next of kin

Audit & Compliance

Full audit logging and structured compliance reporting for regulators

Thank You!

Questions & Answers

GITHUB: GITHUB.COM/GATCHI233/CAREHUB

GITHUB: GITHUB.COM/MAYUFISHA/CAREHUB (FORKED)